A Project Synopsis Submitted for

Bachelor of Technology

Project & Seminar (7SA08)



PROJECT TITLE	**SATYASCORE: An AI Hallucination Detection & Scoring Framework**
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SatyaScore: An AI Hallucination Detection & Scoring Framework

Abstract:

Large Language Models frequently produce hallucinated content- outputs that sound fluent and confident but are factually incorrect or fabricated- a problem that is especially severe for underserved languages like Hindi. Satya-Score addresses this through a claim-level hallucination detection and scoring framework for English and Hindi, with Marathi planned as a future extension. At its core is the Composite Multilingual Hallucination Scoring Algorithm (CMHSA), which fuses semantic similarity, natural language inference probability, and knowledge graph confidence into a single unified score per claim, classifying each into Verified, Uncertain, or Hallucinated. This fusion mechanism- rather than any individual underlying model- is the project's primary novel and patentable contribution.

While claim-level decomposition, multilingual detection, and knowledge-graph grounding have each been explored separately, no existing system combines all three into one fused, real-time-deployable, Hindi-covering framework- the specific gap Satya-Score occupies. The project targets strong F1-scores across English and Hindi, validated through an annotated claim dataset and a knowledge-graph ablation study, with outcomes carried forward through a research publication and an Indian Provisional Patent filing. Marathi support remains a natural next step, aided by existing coverage in models like MuRIL and IndicSBERT.

I. Introduction:

Large Language Models have become part of how we generate and consume text today- powering conversational assistants, automated content generation, and information retrieval systems. But for all their fluency, these models carry a well-known flaw: they hallucinate, producing text that sounds confident and reads perfectly well while being factually wrong, unsupported, or just made up [7]. As LLMs get pulled into domains where factual data is critical - journalism, healthcare information, education, customer-facing applications- catching and quantifying this hallucinated content stops being optional and starts being essential.

This project sits within Natural Language Processing, right at the point where fact verification [1], multilingual NLP [17], and knowledge-graph-grounded reasoning [24] meet. Most hallucination detection work so far has focused almost entirely on English, leaving a real gap for Indian languages like Hindi- despite Hindi having hundreds of millions of speakers and a growing footprint in LLM-generated content [12]. This project, Satya-Score, is our attempt to close that gap: a claim-level hallucination detection and scoring framework built for English and Hindi, with Marathi deliberately marked out as the next language in line.

At the heart of the system is a fusion mechanism we call the Composite Multilingual Hallucination Scoring Algorithm (CMHSA). Most existing approaches lean on just one verification signal [1], [3]. CMHSA instead pulls together three independent signals- semantic similarity, Natural Language Inference (NLI) probability, and Knowledge

Graph (KG) confidence [23], [25]- into one unified score per claim. Each LLM output is first broken down into individual, checkable claims [2], then scored and sorted into one of three tiers: Verified, Uncertain, or Hallucinated.

Marathi, spoken by over 80 million people, feels like the natural next step once the core system is up and running- and resources like MuRIL and IndicNLPSuite already give us a real head start there [18], [19].

a. Motivation :

Large Language Models have found their way into almost every corner of how we generate and consume text today - chatbots, writing tools, search assistants, customer support systems. What makes them so widely adopted is also what makes them risky: they produce content that sounds fluent, confident, and well put together, even when it's factually wrong or backed by nothing real. This is not a rare or occasional failure. Hallucination, as it's commonly called, is baked into how generative models work. And as more people lean on LLM-generated text for education, journalism, customer service, or just everyday questions, the fallout from an unnoticed hallucination becomes harder to ignore which spreads misinformation and gradually reduces user trust in AI-generated content.

This gap is especially hard to ignore when it comes to Hindi. It's spoken by hundreds of millions of people and appears constantly in LLM-generated content now, yet there's still no dependable way for a Hindi-speaking user to check whether a given claim in a response is actually true. And this isn't because the underlying technology doesn't exist - models like XLM-R and MuRIL, and knowledge sources like Wikidata, are all readily available. The required components already exist individually. What is missing is a system that integrates them with genuine Hindi support. This is what motivates the development of Satya-Score.

b. Problem Statement

There's no reliable system that can check individual claims in text generated by LLM in Hindi and English against real facts and flag exactly which ones are hallucinated. Most existing tools only judge an entire response as "reliable" or "not," without pointing to the specific claim that's actually wrong - which makes the feedback far less useful when only one part of a longer response is false. On top of that, almost none of these tools properly support Hindi, leaving a large group of users with no real way to verify what an LLM tells them.

Satya-Score is built to solve this — breaking a response down into individual claims, checking each one against real, structured facts, and doing this reliably for both English and Hindi.

II. Feasibility Study:

The feasibility study for Satya-Score was conducted across four dimensions - technical, time and resource, need, significance.

Technical Feasibility: Satya-Score relies entirely on pre-existing, well-established models and resources - XLM-R, MuRIL, and DeBERTa for semantic and inference-

based scoring, and Wikidata as the structured knowledge source for fact grounding. None of these components need to be built from scratch; the actual engineering effort lies in designing the fusion mechanism (CMHSA) that combines their outputs into a single, coherent score. This significantly lowers technical risk and makes the project achievable within an academic timeframe. The team has structured the work across defined roles - model development, data engineering, backend/pipeline integration, and research/frontend - allowing parallel progress rather than a single bottleneck.

Time and Resource Feasibility: The project is scoped for completion within approximately 15–16 weeks, compressed from an original six-month plan, with clear weekly milestones tied to each team member's role. Required datasets and pretrained models are freely and publicly accessible, and the compute requirements, while non-trivial, fall within the range typically available through academic GPU access or cloud credits, making the project financially and operationally feasible for a final-year team.

Need: Hallucination remains a widely acknowledged, unresolved problem in LLM deployment, and no existing system offers claim-level, knowledge-grounded, multilingual detection with genuine Hindi coverage in one unified framework. This absence directly justifies the need for the proposed system.

Significance: Beyond academic contribution, Satya-Score addresses a practical gap with real-world relevance - enabling more trustworthy LLM deployment for Hindi-speaking users specifically, a segment currently underserved by existing hallucination detection research, while contributing a patentable fusion methodology and a publishable dataset and evaluation framework to the broader research community.

III. Literature Survey:

Hallucination in Large Language Models has drawn substantial research attention over the past few years, and the existing work can be grouped into five broad strands: foundational detection methods, multilingual approaches, cross-lingual embedding resources, fact-verification/RAG techniques, and knowledge-graph-grounded detection. Reviewing all of these together reveals a consistent pattern - strong progress within each strand individually, but very little effort to bring them together.

Foundational Detection Approaches. Early work such as FEVER [1] established the now-standard claim-evidence-verdict pipeline, later refined by FActScore [2] into fine-grained, atomic-claim checking - a principle this project adopts directly for its claim-decomposition step. SelfCheckGPT [3] proposed a zero-resource alternative that checks a model's self-consistency across repeated samples, but this approach fails precisely when a model is confidently and repeatedly wrong, since there is no external source to check against. TRUE [4] provides useful evaluative scaffolding rather than a new detection method.

Benchmarks such as TruthfulQA [5] and HaluEval [6] contributed valuable static datasets but do not address claim-level, multi-signal scoring, while the Huang et al. survey [7] offers a broad framing of the field without proposing a new method itself.

More recent work - RefChecker [8], ANAH [9], and Su et al.'s MIND [11] - pushed toward finer-grained and real-time detection; MIND in particular is relevant to this project's latency goals, though it requires white-box access to a model's internal activations, which is unavailable when auditing closed or third-party LLMs - a constraint CMHSA's fully external, output-only design avoids. Across this entire category, a consistent limitation holds: every method relies on a single verification signal, and none offer meaningful Hindi coverage.

Multilingual Hallucination Detection. This is where the closest language-coverage prior art sits. Mu-SHROOM [12], a large SemEval-2025 shared task, established genuine Hindi-language hallucination benchmarking across 14 languages, and its documented finding of high annotator disagreement directly informed this project's own inter-annotator agreement target. Competing Mu-SHROOM systems - MSA [13], HausaNLP [14], and Hallucination Detectives [15] - explored ensemble voting, model-awareness, and semantic-role-based decomposition respectively, but notably, none of them incorporate any structured knowledge-graph verification.

SHROOM-CAP [16] extended coverage further to nine languages including Hindi, but restricts itself to binary factual/fluency labels within the narrow domain of scientific publications, rather than graded, general-domain scoring. Because of Mu-SHROOM and SHROOM-CAP, this project cannot claim to be the first system offering Hindi-language hallucination detection - that claim has already been established. What remains genuinely absent is the fusion of that language coverage with graded, knowledge-grounded scoring.

Cross-lingual Embeddings and Indic Models. This category consists largely of infrastructure rather than competing prior art. XLM-R [17] and MuRIL [18] provide the multilingual and Indic-specific embedding backbones this project builds on, with MuRIL additionally handling transliterated Hindi text common in real-world queries. IndicNLPSuite [19] supplies evaluation-benchmark conventions this project aligns with.

IndicSBERT [20] and IndicXNLI [21] supply, respectively, sentence-similarity models directly usable for the semantic-similarity component, and a fine-tuning resource for the inference component - both of which also extend to Marathi, supporting the feasibility of the project's future-work extension.

Fact Verification and RAG. Retrieval-Augmented Generation [22] established that grounding a model's output in retrieved evidence at generation time reduces factual errors, and conceptually motivates this project's own knowledge-retrieval design. However, RAG intervenes during generation, whereas this project evaluates already-generated text after the fact - a necessary distinction for a deployed auditing tool that cannot always control or modify how the underlying LLM produces its output.

Knowledge-Graph-Grounded Detection. This is the most critical category for establishing novelty. MultiHal [23] is the single closest prior work, evaluating semantic similarity, inference, and knowledge-graph-based hallucination scores together across five European languages - but critically, it reports these three signals separately rather than fusing them into one combined score, and it does not cover Hindi or offer real-time deployment. FactKG [25] demonstrated that knowledge-graph-based verification

scales reliably across varied reasoning complexity, but uses knowledge-graph checking as its only signal, with no multilingual coverage. KGR [26] is architecturally the closest match to this project's overall pipeline shape, but its goal is autonomous correction of hallucinated text rather than scoring and flagging it for review.

Two independent surveys, Agrawal et al. [24] and Lavrinovics et al. [27], confirm that the question of whether and how knowledge graphs can reduce hallucination remains an open, actively researched problem rather than a solved one.

Identified Gap. Taken together, the reviewed literature reveals two parallel but disconnected research tracks: multilingual hallucination detection systems that lack knowledge-graph grounding (Mu-SHROOM, SHROOM-CAP), and knowledge-graph-grounded systems that lack meaningful multilingual or Hindi coverage (MultiHal, FactKG, KGR). No reviewed system combines claim-level decomposition, a single fused semantic-similarity, inference, and knowledge-graph score, real-time deployability, and Hindi-language coverage within one unified framework. This is the specific, well-defined space that Satya-Score is designed to occupy - The novelty does not lie in language coverage or knowledge-graph use individually, since both are already established in prior work. It lies in fusing these previously separate capabilities into one deployable system.

IV. Objective & Scope of the Project:

The objectives of this project are structured incrementally - moving from understanding the problem, through reviewing existing work, to designing the solution, and finally validating it experimentally.

i) To identify the scope and impact of hallucination in Large Language Models

ii) To analyse existing hallucination detection approaches.

iii) To design and develop the Composite Multilingual Hallucination Scoring Algorith.

iv) To evaluate the developed system through a dedicated ablation study and performance testing on an annotated claim dataset.

Scope: The core deliverable is limited to English and Hindi, with Marathi identified as a future extension. Novelty is scoped specifically to the CMHSA fusion methodology, not the underlying pretrained models (XLM-R, MuRIL, DeBERTa, Wikidata), all of which are pre-existing and publicly available. The current scope ends at detection and scoring - automatic correction of hallucinated content is not included.

V. Requirement:

Software Requirements:

Sr. No.	Category	Requirement
1	Programming Language	Python 3.x
2	Deep Learning Frameworks	PyTorch, Hugging Face Transformers
3	Pretrained Models	XLM-R, MuRIL, DeBERTa-v3-large
4	Knowledge Base / KG Access	Wikidata (via SPARQL endpoint queries)
5	Vector Search / Retrieval	FAISS
6	Backend Framework	FastAPI / Flask
7	Data Handling	Pandas, NumPy
8	Version Control	Git / GitHub
9	Operating System	Windows / Linux (Ubuntu preferred)
10	Development Environment	Jupyter Notebook / VS Code

Hardware Requirements:

Sr. No.	Component	Specification
1	Processor	Intel i5 / i7 or equivalent (minimum)
2	RAM	Minimum 16 GB
3	GPU	NVIDIA CUDA-supported GPU (e.g., RTX-series) or cloud-based GPU access (Google Colab Pro / cloud

		credits)
4	Storage	Minimum 50–100 GB free space
5	Internet Connectivity	Stable, high-speed connection for API access and model/dataset downloads

References:

[1] J. Thorne, A. Vlachos, C. Christodoulopoulos, and A. Mittal, "FEVER: A large-scale dataset for fact extraction and VERification," in Proc. NAACL-HLT, 2018, doi: 10.18653/v1/N18-1074

[2] S. Min et al., "FActScore: Fine-grained atomic evaluation of factual precision in long form text generation," in Proc. EMNLP, 2023. doi: 10.48550/arXiv.2305.14251

[3] P. Manakul, A. Liusie, and M. J. F. Gales, "SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models," in Proc. EMNLP, 2023. doi: 10.48550/arXiv.2303.08896

[4] O. Honovich et al., "TRUE: Re-evaluating factual consistency evaluation," in Proc. NAACL-HLT, 2022. doi: 10.18653/v1/2022.naacl-main.287

[5] S. Lin, J. Hilton, and O. Evans, "TruthfulQA: Measuring how models mimic human falsehoods," in Proc. ACL, 2022. doi: 10.18653/v1/2022.acl-long.229

[6] J. Li, X. Cheng, X. Zhao, J.-Y. Nie, and J.-R. Wen, "HaluEval: A Large-Scale Hallucination Evaluation Benchmark for Large Language Models," in Proc. EMNLP, 2023. doi: 10.18653/v1/2023.emnlp-main.397.

[7] L. Huang et al., "A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions," ACM Transactions on Information Systems, vol. 43, no. 2, 2025. doi: 10.1145/3703155.

[8] X. Hu et al., "Knowledge-Centric Hallucination Detection," in Proc. EMNLP, 2024. doi: 10.18653/v1/2024.emnlp-main.395.

[9] Z. Ji et al., "ANAH: Analytical Annotation of Hallucinations in Large Language Models," in Proc. ACL, 2024. doi: 10.18653/v1/2024.acl-long.442.

[10] A. Mishra, A. Asai, V. Balachandran, Y. Wang, G. Neubig, Y. Tsvetkov, and H. Hajishirzi, "Fine-grained Hallucination Detection and Editing for Language Models," in Proc. COLM, 2024. doi: 10.48550/arXiv.2401.06855.

[11] W. Su, C. Wang, Q. Ai, Y. Hu, Z. Wu, Y. Zhou, and Y. Liu, "Unsupervised Real-Time Hallucination Detection based on the Internal States of Large Language Models," in Findings of the Association for Computational Linguistics: ACL 2024, 2024. doi:

10.18653/v1/2024.findings-acl.854

[12] R. Vázquez et al., "SemEval-2025 Task 3: Mu-SHROOM, the multilingual shared task on hallucinations and related observable overgeneration mistakes," in Proc. SemEval, 2025. identifier: https://aclanthology.org/2025.semeval-1.1

[13] B. Hikal, A. Nasreldin, and A. Hamdi, "MSA at SemEval-2025 Task 3: High Quality Weak Labeling and LLM Ensemble Verification for Multilingual Hallucination Detection," in Proc. 19th International Workshop on Semantic Evaluation (SemEval-2025), 2025. identifier: https://aclanthology.org/2025.semeval-1.131/

[14] M. Bala, A. Abubakar, A. Abubakar, A. Bichi, H. Ahmad, S. A. Sani, I. Abdulmumin, S. H. Muhammad, and I. S. Ahmad, "HausaNLP at SemEval-2025 Task 3: Towards a Fine-Grained Model-Aware Hallucination Detection," in Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025), 2025. identifier: https://aclanthology.org/2025.semeval-1.227/

[15] P. Elchafei and M. Abu-Elkheir, "Hallucination Detectives at SemEval-2025 Task 3: Span-Level Hallucination Detection for LLM-Generated Answers," in Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025), 2025. identifier: https://aclanthology.org/2025.semeval-1.84/

[16] A. Sinha, R. Vázquez, T. Mickus, R. Agarwal, I. Buhnila, P. Schmidtová, F. Gamba, D. K. Prasad, and J. Tiedemann, "SHROOM-CAP: Shared task on hallucinations and related observable overgeneration mistakes in crosslingual analyses of publications," in Proc. CHOMPS, 2025. doi: 10.18653/v1/2025.chomps-main.7

[17] A. Conneau et al., "Unsupervised cross-lingual representation learning at scale," in Proc. ACL, 2020. DOI: 10.18653/v1/2020.acl-main.747 [18] S. Khanuja et al., "MuRIL: Multilingual representations for Indian languages," 2021. doi: 10.48550/arXiv.2103.10730

[18] S. Khanuja et al., "MuRIL: Multilingual representations for Indian languages," 2021. doi: 10.48550/arXiv.2103.10730

[19] D. Kakwani, A. Kunchukuttan, S. Golla, G. N.C., A. Bhattacharyya, M. M. Khapra, and P. Kumar, "IndicNLPSuite: Monolingual corpora, evaluation benchmarks and pre-trained multilingual language models for Indian languages," in Findings of EMNLP, 2020. doi: 10.18653/v1/2020.findings-emnlp.445

[20] S. Deode et al., "L3Cube-IndicSBERT: A Simple Approach for Learning Cross-lingual Sentence Representations Using Multilingual BERT," 2023. doi: 10.48550/arXiv.2304.11434.

[21] D. Aggarwal, V. Gupta, and A. Kunchukuttan, "IndicXNLI: Evaluating Multilingual Inference for Indian Languages," arXiv:2204.08776, 2022. doi: 10.48550/arXiv.2204.08776

[22] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," arXiv:2005.11401, 2020. doi: 10.48550/arXiv.2005.11401

[23] K. Lavrinovics, R. Biswas, K. Hose, and J. Bjerva, "MultiHal: Multilingual dataset

for knowledge-graph grounded evaluation of LLM hallucinations," arXiv:2505.14101, 2025. doi: 10.48550/arXiv.2505.14101

[24] G. Agrawal et al., "Can Knowledge Graphs Reduce Hallucinations in Large Language Models? A Survey," arXiv:2311.09758, 2024. doi: 10.48550/arXiv.2311.09758.

[25] J. Kim, S. Park, Y. Kwon, Y. Jo, J. Thorne, and E. Choi, "FactKG: Fact verification via reasoning on knowledge graphs," in Proc. ACL, 2023. doi:10.18653/v1/2023.acl-long.895

[26] X. Guan et al., "Mitigating Large Language Model Hallucinations via Autonomous Knowledge Graph-Based Retrofitting," in Proc. AAAI, vol. 38, no. 16, pp. 18126–18134, 2024. doi: 10.1609/AAAI.V38I16.29770.

[27] K. Lavrinovics, R. Biswas, J. Bjerva, and K. Hose, "Knowledge Graphs, Large Language Models, and Hallucinations: An NLP Perspective," Journal of Web Semantics, vol. 85, art. 100844, 2025. doi: 10.1016/j.websem.2024.100844.

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